PAIR PROJECT PROPOSAL

Traffic Intensity and Road Maintenance Need in England, 2008–2025

1.Research Question and Basic Information

To what extent is road traffic intensity associated with the proportion of local A-road and local-authority-managed motorway length classified as red across English local authorities between FYE 2008 and 2025?

Unit of analysis. The local-authority-year. Each observation pairs one authority with one financial year ending (FYE).

Course	CISC7204 Data Analysis and Visualization	**Region**	England, United Kingdom
Team	Member 1: [ZHU QINYI / MC648900] Member 2: [XU XIAOYAN / MC647246]	**Domain**	Transportation – Road Infrastructure

2. Project Purpose

Why this topic matters. This project uses two official UK transport datasets to study the relationship between traffic intensity and road maintenance need across English local authorities. The project gives us a practical setting to apply the data analysis and visualization methods covered in CISC7204.

Outcomes. **Outcomes.** The project has produced a cleaned and integrated dataset containing 2,570 observations from 158 English local authorities between 2008 and 2025. We have examined traffic intensity and road maintenance need through descriptive statistics, visualizations, correlation analysis, and regression models. The initial findings show a weak positive association between the two variables. We have also evaluated different models to examine how well traffic intensity, year, and regional information explain maintenance needs. This project allows us to apply the methods learned in CISC7204 to real public-sector data and develop a better understanding of the strengths and limitations of data-driven analysis.

Skills. We will practise data cleaning, reshaping, feature engineering, and dataset merging with Python. We will use pandas and NumPy to prepare the data and create the traffic-intensity variable. We will use Matplotlib and Seaborn to explore distributions, time trends, regional differences, and relationships between variables. We will also use correlation and regression methods to examine the research question. We will compare model performance with training and testing data when appropriate.

Knowledge. The project will strengthen our understanding of the main stages of a data science workflow. We will learn how data types, missing values, variable definitions, and dataset structure can affect an analysis. We will also develop our understanding of exploratory data analysis, descriptive statistics, correlation, and regression. The project will help us choose suitable visualizations for different types of variables and analytical questions. We will also learn how to interpret patterns, outliers, fitted relationships, and model evaluation results. We will distinguish statistical association from causation when we interpret the findings.

3. Data Sources and Feasibility

Feasibility summary: Both datasets were obtained from official Department for Transport sources and cover the common analysis period of 2008–2025. After cleaning and merging the data using local-authority codes and years, the resulting dataset contains 2,570 valid observations from 158 English local authorities. This confirms that the available data support the proposed analysis.

Dataset	Coverage / variables	Role in project	Official source
RDC0120 – Local-authority-managed classified road where maintenance should have been considered (red)	England; FYE 2008–2025. Main outcome: percentage of local A-road and local-authority motorway length categorised red.	Outcome variable, and comparison across regions and years.	RDC0120 ODS file – Road condition statistics: data tables (RDC), GOV.UK
Local authority traffic by vehicle class	Great Britain; 1993–2025. Local-authority code, year, road length, cars and taxis, and all motor vehicles (vehicle-miles).	Used to construct the traffic-intensity predictor and to align it with the road-condition observations.	Local authority traffic CSV – Road traffic statistics: bulk downloads, Department for Transport

4. Project Tasks and Methodology

1) **Data acquisition and audit:** The team will import the two official datasets and inspect their structure, variables, units, geographic coverage, and time periods. We will also identify missing-value conventions and relevant administrative notes.
2) **Road-condition cleaning and reshaping:** The team will use the RDC0120 A-roads and motorways worksheet. Values marked [x], [y], and [z] will be treated as unavailable or non-comparable rather than as zero. Aggregate rows will be removed, labels will be cleaned, and the financial-year columns will be reshaped from wide to long format.
3) **Traffic filtering and feature engineering:** The traffic dataset will be filtered to England and to the common period from 2008 to 2025. Traffic intensity will be calculated as annual motor-vehicle traffic divided by road-network length, giving annual vehicle-miles per road-mile.
4) I**ntegration:** The two datasets will be merged using the ONS local-authority code and year. The team will check code changes caused by local-authority reorganisation before merging records across time.
5) **Exploratory analysis:** The analysis will examine distributions, missing values, outliers, annual trends, and regional differences. Scatter plots, time-series plots, and regional comparisons will be used to identify patterns. Annual averages will be reported as unweighted means across available local authorities.
6) **Association and modelling:** Pearson correlation was used to examine the relationship between traffic intensity and maintenance need. We also calculated the correlation using local-authority averages to check whether the relationship remained after reducing repeated yearly observations. Simple linear regression, multiple linear regression, and

polynomial regression were developed to compare their ability to explain variations in maintenance needs. The multiple regression model included traffic intensity, year, and regional indicators.

7) **Evaluation and robustness:** The dataset was divided into training (80%) and testing (20%) subsets. Model performance was evaluated using R^2 and mean squared error (MSE). Four-fold cross-validation was conducted to examine the stability of the results. We also tested Ridge regression with different regularization parameters. A temporal holdout was performed by training the model on data from 2008–2021 and testing it on data from 2022–2025 to examine its performance on later years.
8) **Interpretation and reporting.** The results were interpreted using correlation coefficients, model evaluation metrics, and supporting visualizations. We compared the explanatory power of traffic intensity alone with that of models including year and regional differences. The analysis also considered limitations involving road classifications, missing data, differences between calendar and financial years, and other unobserved factors. The findings were interpreted as statistical associations rather than causal relationships.

Primary predictor	Traffic intensity = annual all-motor-vehicle traffic (vehicle-miles) ÷ road-network length (miles).
Primary outcome	Percentage of local A-road and local-authority motorway length categorised red, that is, length that should have been considered for maintenance.

5. Visualization, Analysis and Justification

Figure 1 illustrates the relationship between traffic intensity and road maintenance need across English local authorities from 2008 to 2025. Each point represents one local authority in one year. The horizontal axis measures annual vehicle-miles per road-mile, while the vertical axis shows the percentage of local A-road and motorway length requiring maintenance consideration. The fitted regression line indicates the overall direction of the association, while the distribution of points reveals substantial variation among authorities.

The analysis identifies a weak positive correlation between traffic intensity and maintenance need ($r = 0.142$, $p < 0.001$). After averaging observations within each local authority, the correlation remains positive ($r = 0.227$, $p = 0.004$). However, the wide dispersion of points suggests that traffic intensity alone cannot adequately explain maintenance needs. Further analysis shows regional differences, with London recording the highest average maintenance percentage and the North East the lowest. These findings justify the inclusion of year and regional variables in the regression analysis. The visualizations therefore help identify patterns, compare differences, and assess whether the observed relationship is consistent across the dataset.

Model Evaluation Results: The multiple linear regression model achieved the best performance on the randomly selected test set, with $R^2 = 0.1719$ and MSE = 11.0088. In comparison, the simple linear regression model using traffic intensity alone produced $R^2 = -0.0027$. Polynomial regression did not provide a meaningful improvement. Four-fold cross-validation gave an average R^2 of 0.074, while the temporal holdout test produced $R^2 = -$

0.0325. These results indicate that year and regional information improve explanatory performance, but the overall predictive ability remains limited. Traffic intensity alone is therefore insufficient to explain variations in road maintenance needs.



Figure 1. Relationship between traffic intensity and road maintenance need across English local authorities, 2008–2025.

Dataset links. The two official datasets used in this project are available from the following sources. 1. Road condition statistics: data tables (RDC), GOV.UK. 2. Road traffic statistics: bulk downloads, Department for Transport. The analysis dataset derived from these two sources is exported by the notebook as england_a_road_clean.csv.

Verification notebook . The file CISC7204_Traffic_Road_Condition_Project.ipynb reproduces every figure and every statistic cited in this proposal, and it runs under Python 3.

6. Criteria for Success

- The project should successfully clean and integrate the two datasets at the local-authority-year level. The main data-processing decisions should be clearly documented.
- The project should define the traffic-intensity measure and road-condition outcome clearly and consistently.
- The analysis should use suitable statistical methods and visualizations to address the research question.
- The project should evaluate the reliability of the results with appropriate model evaluation methods and should consider the structure and limitations of the data.
- The final interpretation should be supported by the analysis. The project will be considered successful whether the relationship is strong, weak, or absent, as long as the findings are reported accurately and the main limitations are clearly explained.

Evaluation criteria. The project will be assessed based on the reproducibility of data processing, the clarity of visualizations, the appropriateness of statistical methods, and the interpretation of the findings. Model performance will be evaluated using test-set R^2, MSE, cross-validation, and temporal holdout results. A weak statistical relationship or low predictive accuracy will not be considered a failure if the findings are accurately reported and the limitations are clearly explained.

Planned deliverables. The project deliverables include a cleaned and merged CSV dataset, a reproducible Python notebook, clearly labelled visualizations, statistical analysis results, model comparisons, and a written interpretation of the findings.

7. Collaboration Plan

Role	Primary responsibilities	Cross-review / shared responsibility
Member A – Data engineering lead	Import datasets; inspect data structure; clean missing values; reshape data; create variables; merge datasets by local authority and year.	Member B reviews the cleaning steps, variable definitions, and merged dataset.
Member B – Analysis and visualisation lead	Exploratory analysis; core and supporting figures; correlation and regression; model evaluation; interpretation and figure captions.	Member A reviews model inputs, figures, labels, and interpretation of results.
Shared	Refine the research question; discuss limitations; interpret findings; prepare the proposal and final submission.	Both members check the notebook, figures, conclusions, and final document before submission.

Ground rules. Both members will keep the raw data unchanged and use shared files for all project work. Each member will review the other member's work before a section is completed. The team will record important analytical decisions in the notebook. Both members will discuss any disagreement by referring to the research question and the original data definitions.

Team structure. Member A will act as the project coordinator and manage the project schedule. Member B will act as the project scribe and maintain the notebook records. Both members will share the progress tracker role. Member B will act as the project liaison for communication with the teaching team.

Use of Generative AI

We acknowledge the use of **ChatGPT (OpenAI)** to support background research and independent study, to generate materials that were reviewed and adapted for this proposal, and to assist with copy-editing and proofreading. ChatGPT was also used to clarify data science concepts, support Python debugging, and discuss data-cleaning and visualization methods. The team independently conducted and verified the data processing, analysis, visualizations, and model evaluation. All AI-assisted material was critically reviewed and revised before inclusion. The team takes full responsibility for the accuracy, interpretation, academic integrity, and final content of this submission.